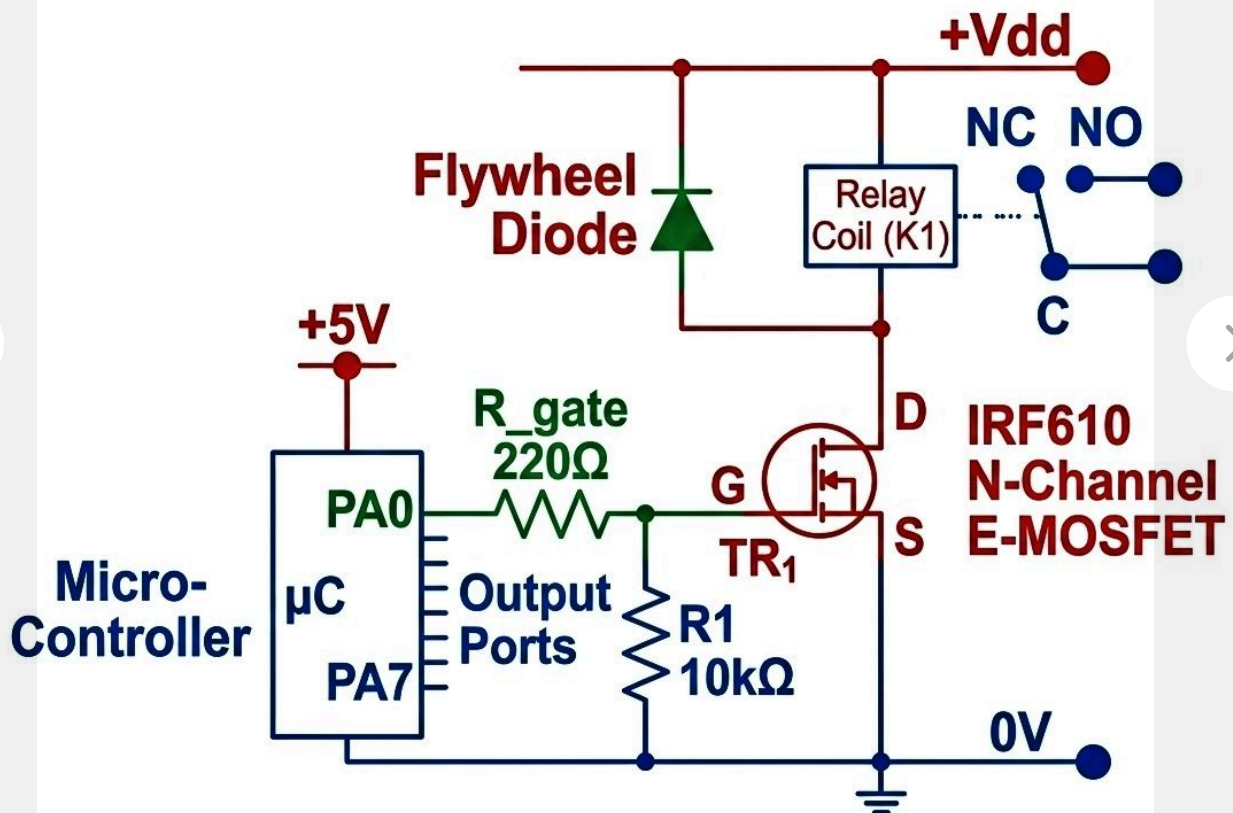




MICROCONTROLLER RELAY CONTROL



Electronics Education

2 maj kl. 17:49 · 🌐

Controlling a relay directly from a microcontroller pin is not safe because the relay coil needs more current than the pin can supply. This circuit solves that using an N-channel MOSFET (IRF610) as a switch. When the microcontroller output (PA0) goes HIGH, a small current flows through the 220Ω gate resistor, turning the MOSFET ON. This allows current to flow from +Vdd through the relay coil to ground, energizing it and switching the contacts from NC to NO.

When the output goes LOW, the MOSFET turns OFF and the relay deactivates. The 10kΩ resistor ensures the MOSFET stays OFF when no signal is present. The flyback diode is critical—it protects the MOSFET from high voltage spikes generated when the relay coil turns OFF.

This is a standard and reliable method used in embedded systems to safely control high-power loads from low-power logic signals.

#ElectronicsEducation #ElectronicsRD #EmbeddedSystems #Microcontroller #MOSFET #STEM #STEMEducation #STEMLearning Visa mindre

👍 301 💬 10 ➦ 45

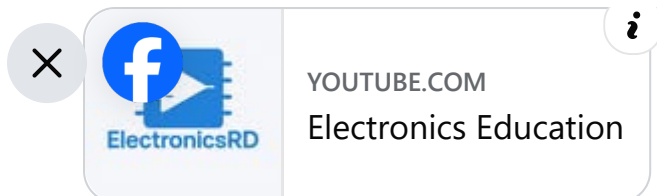
Mest relevant ▼



Författare

Electronics Education

Youtube: https://youtube.com/@electronicseducation?sub_confirmation=1... Visa mer



2 v **Gilla** Svvara



Branislav Kysel

To direct control the mosfet from microcontroler, you need to use logic level mosfet. Something like e.g. IRL 5X0 series. It w... **Visa mer**

2 v **Gilla** Svvara Redigerad 2



Christo van der Merwe

Suggestion, a general purpose BJT will do the same but is cheaper and not sensitive to static electricity.

2 v **Gilla** Svvara 1

Visa 1 svar



David Simons

IRF610 has a pretty high Vgs at room temperature. Might not switch much current without a booster. Consider the venerable irlz... **Visa mer**

1 v **Gilla** Svvara



Massimo Decimo Meridio

En eldåd för att driva en relä?

2 v **Gilla** Svvara Visa ursprungsversion (Italienska)



Stort fan

Eko Wahyudi

Saya mau tanya kalau Com Relay Negatif gimana Weiring. Terimakasih bila ada Jawabannya

2 v **Gilla** Svvara

Visa 4 kommentarer till

6 av 8



Skriv en kommentar ...

